



Implementing Manufacturing Execution Systems (MES) for Industry 4.0: Overcoming buyer-provider information asymmetries through knowledge sharing dynamics

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ARTICLE INFO

Keywords:

Industry 4.0
Manufacturing Execution System
Digital transformation
Information asymmetry
Knowledge Sharing

ABSTRACT

Manufacturing Execution Systems (MES) have been considered the ‘entrance door’ to the Industry 4.0 journey. As part of this journey, this system must operate in real-time and be integrated with several other systems, resulting in modular adaptation and customized implementation of this solution. This increased complexity underscores the importance of the relationship between companies and their technology providers, requiring intensive knowledge-sharing (KS) activities between the parties. In particular, information asymmetry between buyers and MES 4.0 providers may be critical for the successful implementation of the system, but little is known about this issue, which has a high impact on the first stage of the Industry 4.0 journey. Thus, we aim to understand how knowledge sharing affects information asymmetry between buyers and technology providers for MES 4.0 implementation. Our investigation is based on qualitative interviews with 56 key experts from 33 companies, which allowed us first to define MES configurations for Industry 4.0, i.e., MES 4.0 as a differentiation from early versions of MES. Then, we conducted a multiple case study with three buyer-provider dyads in the pre-and post-contract phases of MES purchasing to analyze KS dynamics during the implementation of MES 4.0. We propose a model that explains the relationship between MES 4.0 complexity and KS intensity and a decision framework that describes the steps for MES 4.0 implementation while focusing on reducing information asymmetry during the collaboration.

1. Introduction

The extensive array of new applications of digital technologies in industrial activities has presented numerous opportunities for companies to embark on the Industry 4.0 journey—a technological paradigm aimed at progressively implementing interrelated digital technologies to enhance operational performance, including productivity, quality, flexibility (Enrique et al., 2022). The Industry 4.0 journey towards a digitally-enabled factory begins with enhancing visibility capabilities on the shop floor (Benitez et al., 2023). This capability enables companies to follow real-time events in the factories and integrate this information with others to facilitate better decision-making (Dalenogare et al.,

2018).

Manufacturing Execution Systems (MES) play a pivotal role in providing visibility, as they are directly linked to the operational structure of the shop floor, encompassing machines, equipment, sensors, Programmable Logic Controllers (PLCs), and Supervisory Control and Data Acquisition (SCADA) systems (Pérez-Lara et al., 2018; Tamas et al., 2019). By collecting and integrating data from these elements, MES facilitates data management and the recording of production data history, crucial for informed decision-making (Tabim et al., 2021). Moreover, as Industry 4.0 matures, more advanced levels such as autonomous systems emerge, where machines within production lines autonomously make decisions based on real-time data (Tariq et al., 2024). This needs

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<https://doi.org/10.1016/j.cie.2024.110483>

Received 4 October 2023; Received in revised form 13 May 2024; Accepted 13 August 2024

Available online 22 August 2024

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vertically integrated systems capable of integrating hierarchical data from the shop floor, collected by MES, into intermediate and upper management levels, commonly referred to as the traditional Enterprise Resource Planning (ERP) corporate system (Wang et al., 2023). Consequently, MES emerges as a central platform for connecting various technologies within a smart manufacturing environment (Benitez et al., 2023). Thus, there is a growing interest in implementing and utilizing MES more effectively to achieve real-time visibility and vertical integration. Despite MES's existence since the 1990s (MESA, 2011)—two decades before the term Industry 4.0 was coined—implementing MES with Industry 4.0 requirements (MES 4.0) poses new challenges due to the need for customization to suit the characteristics of each production process (Tariq et al., 2024). Achieving real-time visibility downstream requires integration with shop floor equipment, which typically comprises a mix of old and new equipment employing varying technologies (Tabim et al., 2021). Conversely, Industry 4.0 vertical integration upstream requires connectivity with ERP and other systems, yet many companies still rely on older legacy systems (Benitez et al., 2023).

Given the multifaceted functionalities of the new MES 4.0 (Tariq et al., 2024), effective information and knowledge sharing between technology providers and buyers are demanded (Tabim et al., 2021). Information asymmetry can lead to discrepancies between the two parties, resulting in implementation difficulties, usability issues, and suboptimal system performance (Chattopadhyay & Aundhe, 2021). Technology providers offer MES solutions as Industry 4.0 technologies without fully elucidating their capabilities, limitations, and integration complexities with other systems. Conversely, buyers tend to overestimate the ease of integration and the scope of automated solutions offered by MES. Consequently, asymmetrical information during the purchasing process can culminate in wrong implementation, affecting the company's performance, eroding trust and damaging the reputation of technology providers (Chattopadhyay & Aundhe, 2021). Therefore, successful MES implementation in the context of Industry 4.0 requires a level of knowledge sharing that goes beyond just technology. Due to the system's complexity and multiple functionalities, companies need a certain level of Knowledge Sharing (KS) to prevent information asymmetry during system negotiations and purchases (Carlile, 2004; Tabim et al., 2021). For example, a simple MES 4.0 implementation that only requires production scheduling without real-time upstream and downstream integration would need less KS than a full MES 4.0 integration operating as a platform with all shop floor equipment and ERP systems interconnected (Benitez et al., 2023). Recent literature underscores MES as the primary and indispensable step in the Industry 4.0 journey (Tabim et al., 2021) and as the central platform or "brain" for integrating operational technologies to establish smart manufacturing (Benitez et al., 2023). Nevertheless, little attention has been paid to the challenges associated with integrating different levels of solutions through interactions between buyers and providers. Hence, the following research question is proposed: *How can KS activities between buyers (manufacturers) and MES technology providers reduce information asymmetry between parties when pursuing different levels of solutions in the Industry 4.0 context?*

Our objective is to propose a framework that determines the type of KS necessary to reduce information asymmetry between buyers and technology providers for MES implementation in an Industry 4.0 context based on the system's functionalities. We first apply the information asymmetry perspective (Akerlof, 1978) and combine it with the KS dynamics (Carlile, 2004) to identify the type of KS and related dynamics that would reduce information asymmetry problems, leading to a successful MES implementation project. With this theoretical framework, we conduct qualitative interviews with 56 key experts from 33 companies to identify MES configurations for Industry 4.0. We then conduct a multiple case study with three buyer-provider dyads in the pre- and post-contract phases of MES purchasing to analyze the KS dynamics. Our results indicate that MES 4.0 can have different combinations of functionalities, which depend on the firms' Industry 4.0 maturity and

strategic objectives. Accordingly, the innovativeness and complexity of MES vary, demanding different KS dynamics between buyer and provider that can evolve from simple knowledge transfer to knowledge translation and finally to the highest complexity of knowledge transformation. Moreover, these different types of KS dynamics will require different buyer-provider collaboration configurations (Petersen et al., 2003). This study contributes to the scarce academic literature on the topic by explaining the nuances of buyer-provider relationships for MES implementation in the Industry 4.0 era. It also provides practical guidance for practitioners wanting to advance their Industry 4.0 journey.

2. Theoretical background

2.1. Manufacturing execution system (MES) for Industry 4.0

The Manufacturing Execution System (MES) concept was coined in 1990 to unify the information system layers operating on the shop floor, including machinery, equipment, sensors, PLC, and SCADA, with the ERP system of the traditional automation pyramid (Mantravadi et al., 2023; Chen & Voigt, 2020). In 1992, the Manufacturing Execution Systems Association (MESA) was created. The association created the MESA model, which considered a set of 11 functionalities necessary for an information system to be classified as an MES. These functionalities include (1) resource allocation and status, (2) operations/detail scheduling, (3) dispatching production units, (4) document control, (5) data collection/acquisition, (6) labor management, (7) quality management, (8) process management, (9) maintenance management, (10) product tracking, and (11) genealogy and performance analysis. The MESA initiative aimed to fill the gap in the MES layer with all the potential functionalities that could be implemented for commercial purposes. However, many scholars claim that this definition of MES needs to be more specific, making it difficult to understand the main objective of the MES solution and confusing successful implementation within companies. This scenario worsened after the emergence of Industry 4.0, which put the MES in the spotlight due to its key role in vertical integration (Benitez et al., 2022; Tabim et al., 2021).

In the context of Industry 4.0, MES 4.0 utilizes basic technologies such as the Internet of Things (IoT), Big Data, and Cloud Computing to obtain vertical integration and historical production data in real-time (Jaskó et al., 2020). Therefore, the MESA functionalities should consider data gathering and processing within this new real-time premise. It is expected for the MES 4.0 system to be a protagonist in integrating the Information Technology (IT) and Operations Technology (OT) layers, making its implementation even more complex. However, the definition of concepts, standards, and specifications for integrating Industry 4.0 technologies with MES is still under development, although the Industry 4.0 concept has been around for over a decade. Tabim et al. (2021) showed that adopting MES for vertical integration is a challenging decision that involves critical factors for the success of the implementation. Defining and purchasing an MES in this new scenario requires buyers and technology providers to be aligned with the system's concepts, expectations, and purposes, as it will become a manufacturing platform (Benitez et al., 2023). Therefore, the existence of any *information asymmetry* between the buyer and the technology provider during the process of MES purchasing and implementation could harm both firms and become a barrier to the advancement of Industry 4.0 maturity of the buyer.

2.2. Asymmetry of information and knowledge sharing activities

Aligning the decisions of different actors, such as buyers and providers, who have independent objectives, is a critical challenge (Shao et al., 2021; Ayala et al., 2019). Industry 4.0 has brought even more difficulties to this context due to the need to integrate multiple technologies. This complexity demands specific technical knowledge from the actors, which can increase the information asymmetry between them

(Vosooghizajzi et al., 2020). Information asymmetry is a concept rooted in economics (Akerlof, 1978). In the case of a buyer-provider relationship, it assumes that buyers are at a disadvantage when they cannot distinguish between high- and low-quality products during the purchasing process. This purchasing transaction can be separated into pre- and post-contract stages, and information asymmetries may be present in both (Zavolokina et al., 2021).

Asymmetric information is present in buyer-provider relationships for various reasons, such as the fear of losing competitive advantage, gaining extra benefits or bargaining power, ensuring information system compatibility, and difficulty in expressing ideas (Vosooghizajzi et al., 2020). Information asymmetry is also widely studied in the field of information systems (Fernández-Barcala et al., 2010; Notheisen et al., 2017; Preikschat et al., 2021; Son et al., 2021; Woods & Simpson, 2018; Zavolokina et al., 2021) where product quality assessment is challenging due to the inability to examine products (Zavolokina et al., 2021) physically. The complexity of assessing information quality is further amplified in an Industry 4.0 environment that requires heterogeneous integrated technologies. One example of this complexity is the buyer-provider relationship during the purchasing process of a Manufacturing Execution System (MES) compatible with the vertical integration requirements (Büchi et al., 2020; Dalenogare et al., 2018; Tabim et al., 2021). Tabim et al. (2021) identified several factors in the adoption of systems for vertical integration in Industry 4.0 that must be considered for successful implementation in this new technological scenario. They also reported numerous cases of MES implementation failure due to wrong purchasing decisions resulting from information asymmetry.

A common issue in the market is that providers may not fully disclose the challenges and limitations of MES regarding its implementation and effectiveness (Tabim et al., 2021). Therefore, it is crucial to establish a collaborative relationship between technology providers and buyers, where information and knowledge are shared and processed to minimize information asymmetry (Vosooghizajzi et al., 2020). By exploring information asymmetry, it is possible to clarify the implementation of an MES that allows vertical integration in Industry 4.0. Thus, based on the findings of Tabim et al. (2021), it is evident that this is not solely a technological problem but also an issue of information and knowledge sharing.

To address the challenge of information asymmetry, we propose to use the knowledge-sharing framework introduced by Carlile (2002), which suggests three types of knowledge sharing – transferring, translating, and transforming – to facilitate knowledge exchange between actors and create different dynamics of knowledge sharing (see Fig. 1). Previous studies, such as Le Dain and Merminod (2014) and Ayala et al. (2017), have applied this framework to analyze buyer-supplier relationships, and we use their insights to analyze the process of purchasing and implementing an MES system for vertical systems integration in Industry 4.0. In this context, knowledge sharing is viewed as a process of moving knowledge from a source to a recipient, followed by absorption and use, building on previous experience (Ayala et al., 2017). In our case, the MES providers possess knowledge about the system's functionalities, while the manufacturing companies seek to purchase the MES to align with their Industry 4.0 strategy. Fig. 1 illustrates the three levels of knowledge-sharing complexity across the boundaries between the buyer and the provider. As the parties engage in a digital transformation through the implementation of the MES, the complexity of knowledge sharing grows from knowledge transfer to knowledge translation and, finally, to knowledge transformation. Depending on the MES's complexity in terms of functionalities needed for the Industry 4.0 journey, the actors may exhibit different knowledge-sharing dynamics, moving up or down across the levels of cross-knowledge complexity.

At the level of *knowledge transfer*, knowledge is considered to be external, explicit, and storable. The primary concern at this level is the syntactic capability required to develop a common lexicon that can be

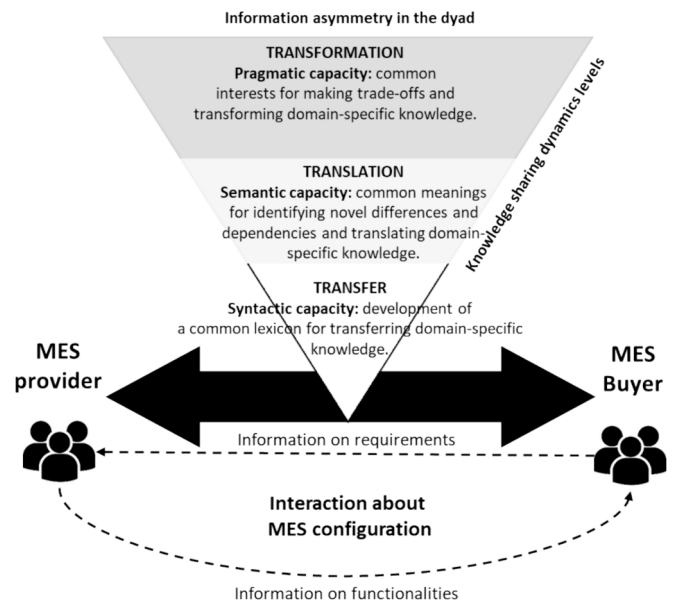


Fig. 1. Knowledge sharing dynamics between buyer and provider (Adapted from Ayala et al., 2017)

used to facilitate communication across boundaries. However, as the level of innovation increases, transferring knowledge may no longer be sufficient because the existing lexicon may no longer be adequate to represent differences and dependencies (Carlile, 2004). In practice, knowledge transfer can be identified when boundary objects are exchanged between actors. Boundary objects are objects or documents created and used during communication, such as requirements, prototypes, design drawings, information on a website, and emails (Carlile, 2004; Le Dain and Merminod, 2014).

At the level of *knowledge translation*, as innovation (in our case, digital transformation) and complexity increase, so does the complexity of knowledge sharing. Thus, transitioning from a syntactic boundary to a semantic or interpretative boundary becomes necessary, as some differences and dependencies become obscure, and meanings can become ambiguous. At this level, the complexity of knowledge sharing leads to different interpretations, requiring mechanisms to create shared meaning between the buyer and the provider to cross this new frontier. Creating common meanings can become more than just a translation process; it can be a process of negotiating interests between actors that may result in learning costs for some of them (Carlile, 2004; Le Dain and Merminod, 2014).

In the highest level of innovation and complexity, the knowledge transformation level, there is a pragmatic frontier that replaces the semantic frontier. At this stage, the divergent interests and realities of the actors need to be resolved. Actors need to learn to accept new knowledge and transform their domain-specific and common knowledge to effectively share and evaluate knowledge across borders (Carlile, 2004). This level poses the most complex frontier, as the cost of transforming existing knowledge can negatively impact the actor's willingness to make the necessary changes (Le Dain and Merminod, 2014).

Given the innovative nature of MES implementation in the context of Industry 4.0 and the varying degrees of complexity of MES 4.0 solutions in the market (Tabim et al., 2021), it is essential to prioritize knowledge management to advance in Industry 4.0 (de Bem Machado et al., 2022; Fakhhar Manesh et al., 2021). Different levels of knowledge sharing dynamics may be necessary to avoid information asymmetry between buyers and providers. Several authors have highlighted the importance of such dynamics, including Le Dain and Merminod (2014) for new product development and Ayala et al. (2017; 2021) for service innovation. To achieve the desired output from an innovative project, buyers

and providers must collaborate in different configurations, as proposed by Petersen et al. (2003): White Box, where the buyer drives solution design; Grey Box, where there is joint solution design; and Black Box, where the provider drives solution design. We propose that combining these perspectives could shed light on the MES purchasing and implementation processes toward Industry 4.0, as discussed in the following sections.

3. Research method

Our research method consisted of three steps. Firstly, we developed a conceptual framework using the perspective of information asymmetry (Akerlof, 1978) and knowledge-sharing theory (Carlile, 2004). We used them to elucidate which knowledge-sharing dynamic could effectively reduce information asymmetry problems, leading to a successful MES implementation project. Secondly, we interviewed 56 key experts from 33 companies that went through the purchase and implementation of MES to understand how specific MES functionalities and characteristics could relate to knowledge-sharing dynamics. Lastly, we analyzed the problem through three in-depth case studies of MES purchasing and implementation, covering both pre- and post-contract phases, and each related to a different level of knowledge-sharing dynamics. We employed a multiple case study approach utilizing qualitative data collection and analysis. We chose this approach to enhance external validity and mitigate potential observer bias (Voss et al., 2002). We followed the guidelines of Voss et al. (2002) for case study research, as described in the following subsections. Additionally, we adopted the methodological procedures of previous works by Benitez et al. (2020) and Son et al. (2021), which analyzed technological and Industry 4.0 environments. Our unit of analysis was the buyer-provider relationship. To capture the two-fold behavior, Galletta (2013) recommends using semi-structured interviews. Therefore, we interviewed buyers and providers involved in each MES implementation. Our analysis focused on the collaboration configuration between the buyer and provider, using knowledge sharing theory (Carlile, 2004) and the framework proposed by Le Dain and Merminod (2014) for the MES purchasing and implementing processes.

3.1. Interview with experts

To gain a deeper understanding of the phenomenon, we conducted semi-structured interviews with 56 key experts from 33 companies that went through the purchase and implementation of MES. Our sample included 27 buyers and nine providers who met our criteria. Table 1 provides a summary of the interviewees' characteristics. Through these interviews, we gained insights into the different MES configurations for the Industry 4.0 journey, which allowed us to pre-select dyads to analyze each of these configurations. Based on the interviews, we identified three types of MES configurations in the Industry 4.0 context that have not been previously described in the literature.

Table 1
Summary of the experts interviewed (n = 56).

Industries	%	Company size	%
Services	30 %	Small (less than 99 employees)	15 %
Automotive	15 %	Medium (between 100 and 500)	39 %
Furniture	12 %	Large (more than 500 employees)	45 %
Beverage	9 %		
Automation/Equipment	9 %		
Electronic	6 %	Job Position	%
Chemical	6 %	Director	23 %
Transportation	3 %	Manager	13 %
Houseware	3 %	Engineer	43 %
Clothes	3 %	Analyst/Specialist	21 %
Agricultural	3 %		

3.2. Multiple case study selection

Based on the insights from the expert interviews, we selected three dyads (three buyers and three providers) using the following criteria: (i) one dyad for each type of MES configuration identified, (ii) a direct relationship between the buyer and provider, without intermediaries, (iii) access to both actors for interviews, and (iv) completion of both pre-contract and post-contract implementation phases. Table 2 provides an overview of each buyer and provider in the studied dyadic relationships.

3.3. Data collection procedures

We first employed semi-structured interviews with MES buyers, and then providers were interviewed. This was necessary to cross the information and match what happened before and after the MES purchasing. The interviews were conducted from September 2020 to October 2021. To achieve the main goal and properly answer the research questions, we used the knowledge sharing perspective to identify information asymmetry problems. We split our analysis into two stages: (i) the first was related to the period before the buyer signed the contract for MES purchasing, and (ii) the second was after the contract.

For the first stage, we considered aspects of information asymmetry, such as when and what providers had more information than the potential buyers and vice-versa. This analysis helped us understand how the dyadic relationship led to information asymmetry and misalignment among providers and buyers. Furthermore, these points supported our knowledge about what buyers understand as MES in Industry 4.0, its functionalities, what providers were selling, and which buyer-provider collaboration configuration was expected.

In the second stage of our analysis, we observed the actual collaboration between buyers and providers in supporting the MES implementation project after the technology was purchased. By examining this stage through a knowledge sharing theoretical lens, we could identify which pieces of information were missing during the MES negotiation process for buyers and how the dyadic relationship between buyers and providers evolved during the implementation phase. This enabled us to draw connections between the different elements and conclude what is necessary for firms to implement MES technology and achieve their Industry 4.0 goals successfully.

Moreover, we contrasted and complemented statements from buyers and providers to understand both parties during MES purchasing before and after the buyer signed the contract. The purpose was also to review and validate evidence, which is important to avoid misinterpretations or bias in reviewers' analyses (Goffin et al., 2019). We performed at least two interviews per dyad (provider and buyer), which lasted around two each. At least two research assistants took notes of the main comments while the main researchers conducted interviews. The interviews were also recorded and later transcribed.

3.4. Data analysis – validity, reliability, and interpretation

To ensure construct validity, we followed Voss et al.'s (2002) recommendation and utilized data triangulation, using multiple sources of evidence within each part of the dyad. Our research consisted of four stages: (1) Identifying buyers and providers; (2) Analyzing dyads' information asymmetry and knowledge sharing before signing the contract, focusing on adverse selection; (3) Integrating and validating data using a data triangulation approach; and (4) Analyzing dyads' information asymmetry and knowledge sharing after signing the contract.

During Stage 1, we aimed to understand how MES technology was dealt with in the market and why buyers purchased it in an Industry 4.0 scenario. We organized all data from interviews to capture how buyers and providers behaved during negotiations of MES technology, utilizing adverse selection from information asymmetry to understand how firms dealt with information during MES negotiations. Stage 2 allowed us to conclude how MES was sold and which functionalities for Industry 4.0

Table 2

Background of the buyers from multiple case study method.

Dyad	Case company	Description	Size	Data Sources	Segment
1	AutomotiveCo	Multinational company from road implements	Multinational (+12,000 employees)	Automation Engineer; Operational Director; Digital Manufacturing Engineer.	Automotive
	DigitalizationCo	Startup from digitalization	Small (+40 employees)	CEO; COO; MES business analyst	Software and IT
2	FurnitureCo	Company from furniture	Large (+1000 employees)	Operational Director; Head of innovation; Industrial Engineer; Process Engineer.	Furniture
	SoftwareCo	Company from automation and software solutions for manufacturing	Small (+10)	Automation Engineer; System Engineer	Software and IT
3	AgriculturalCo	Multinational company from agricultural equipment	Multinational (+20,000 employees)	IT Manager; Manufacturing Manager; Process Engineer	Machinery and equipment
	SystemsCo	Multinational company from industrial systems	Large (+1200 employees)	Industrial Engineer; Business Analyst	Software and IT

were important. In Stage 3, we cross-checked interview data to validate our results, identifying similarities and patterns in the answers. Finally, in Stage 4, we considered the implications of MES purchasing for both sides of the dyadic relationship, analyzing the main results for buyers after purchasing.

To ensure reliability, we utilized independent analyses by all researchers and assistants and discussed differences in codification and interpretation. We also checked for divergences between understandings by crossing data and information between providers and buyers who had a relationship. To validate our insights, we used secondary data such as websites and news and visited all buyers to obtain a deeper understanding of MES implementation.

To ensure external validity, we presented our results in seminars with other researchers and companies with expertise in Industry 4.0 and information systems. These seminars helped us to collect impressions, correct interpretations and discuss the meanings of our findings. Overall, our study employed a rigorous research design and utilized multiple sources of evidence to ensure reliability and construct and external validity.

4. Results and propositions

4.1. Types of MES configuration for Industry 4.0

In the context of Industry 4.0, MES plays a critical role in the information technology infrastructure of manufacturing companies. MES must address the challenges of decentralization, vertical integration, advanced analysis, and cloud computing. The communication paradigm of smart products and features creates new data flows, and integrating these autonomous entities is essential to ensure efficient cooperation between the company and its systems. The Business Director of SoftwareCo2 provided his insights on the role of MES in the Industry 4.0 landscape, stating that “Now MES needs to exchange information in real-time with all hierarchical layers, and no longer act as an independent system forming an information island.” According to the interview with ChemicalCo’s Automation Engineer, joining the IT and OT teams is essential to support the MES 4.0 initiatives. Vertical integration of MES is crucial for the entire system to operate as expected in the Industry 4.0 context. A study by Jaskó et al. (2020) demonstrated how MES can meet new integration requirements and highlighted the current implementation status of these requirements in MES solutions.

A common understanding of the experts interviewed is that the new version of MES in the Industry 4.0 context must provide this real-time data flow. However, more is needed to differentiate from the past MES versions completely. Another thing that MES 4.0 requires is the synchronization of the data collected from the operation with the production planning to evolve toward smart production planning and control (Smart PPC). The interviewees highlighted that few market solutions are today capable of operating like the ‘brain’ of the manufacturing system, as it should be according to the Smart PPC and Smart Manufacturing

view of MES 4.0 (Bueno et al., 2020). However, only those MES designed to be connected to the 4.0 environment will be able to evolve in such a direction. This is why one of the companies highlighted a new technology provider that developed the MES from the ground and has the potential to become much more suitable for the Industry 4.0 journey than other MES 4.0 that resulted from the evolution of prior MES versions.

When considering the MES 4.0 originated from previous systems, the experts pointed out two versions. Some MES evolved from ERP systems to exploit the Internet of Things opportunities. One of the companies interviewed adopted this MES version, a subproduct from the previously acquired ERP. They considered it a MES 4.0 because it runs on an IoT platform, uses cloud systems, and can provide data analyzed with AI algorithms. However, because of its origin from an ERP, the system was more useful for global comparisons between the different factories of the company. In this sense, the MES operated more as a business intelligence (BI) system for monitoring in real-time the multi-plant operations, but was not ruining the shop floor activities. Therefore, we did not focus on this multi-plant profile since our target is those versions oriented to organize the shop floor. A second version is those MES 4.0 focused on shop floor activities, which originated from previous SCADA systems. The experts pointed out that these MES versions are less focused on planning issues, and more oriented to providing real-time monitoring indicators when the operation is running, serving for manufacturing supervision. The more integrated these systems are in the vertical integration structure of the manufacturing informational layers, they can operate as platforms that connect different manufacturing elements and equipment (Benitez et al., 2023).

We summarize these considerations of MES’s evolution in the Industry 4.0 context in Table 3. This table shows three types of MES 4.0 with different levels of requirements according to their evolution. The table describes these three concepts and shows the main cutting-edge digital technologies that are integrated with the MES at each level. We also included some quotations from the experts that highlight how they consider such types of MES 4.0. As some of these experts noted, the first problem in information asymmetry is given due to the lack of understanding of these different levels and types of MES 4.0. One expert from an automotive company said, “We faced several problems with the implementation of our MES because we were expecting something very different from what the vendor proposed. We created our view on the MES for our Industry 4.0 journey, which was not the reality of the products we found in the market”. Expectations and the system available may be different, and this could be reduced when the types of MES 4.0 and their configurations are well understood.

The first configuration of MES 4.0 (Table 3) delivers only local visibility and is usually employed by firms starting their Industry 4.0 journey (Dalenogare, Benitez, Ayala, & Frank, 2018; Frank, Dalenogare, & Ayala, 2019). Firms in this configuration implement specific MES modules, such as data collection, production monitoring (e.g., OEE), product tracking, or quality management (Jaskó et al., 2020). The

Table 3

Types of MES configuration for Industry 4.0.

Types of MES configuration for Industry 4.0	Description	Cutting-edge digital technologies integrated with MES	Example cases
Local visibility	MES delivers only local visibility, such as data collection, production monitoring (e.g., OEE), product tracking, or quality.	IoT systems for real-time data collection. Basic modules of big data analytics for data visibility.	“We expected an MES capable of bringing visibility to a specific line, not yet to the entire plant.” (MetalmechanicsCo) “We bought an MES that was just installed in our production line by our provider.” (ClothesCo)
One-way integration	MES delivers general visibility of the factory. The firms must vertically integrate their systems. The one-way flow of information is enough.	IoT systems for real-time data collection. Cloud systems for integrating data across different processes. Basic modules of big data analytics for data visibility.	“We needed an MES capable of receiving information directly from the ERP system, for example, batch and product expiration information to print on the label.” (CosmeticsCo) “We are working to achieve communication between our MES and the ERP, we still have a way to go.” (BeverageCo)
Two-way integration	MES has a focus on complete vertical integration to leverage digital twins or autonomous processes. The information must have a two-way flow.	IoT systems for real-time data collection. Cloud systems for integrating data across different processes. Big data analytics using AI for autonomous decision-making. Digital twins to conduct simulations and process adaptations based on the real-time data obtained.	“We aim to have an MES that not only sends information to SAP but also receives it, allowing SAP to issue commands to MES.” (TechnologyCo) “We are seeking an MES that has fully integrated IT layer with AT.” (PowertoolsCo)

information is then visualized on dashboards to trigger continuous improvement activities such as root cause analysis, control charts, and PDCA cycles (Telukdarie et al., 2018). These functionalities are usually reached by plug-and-play IoT solutions, with a simple installation in which the sensor data is sent to the cloud and visualized in online software provided by the same provider. Basic modules of big data analytics for data visibility can also be present.

The second configuration of MES 4.0 delivers general factory visibility (Table 3). To reach this level, the firms must vertically integrate their systems through the different layers: IoT sensors, MES, and ERP (Tamas et al., 2019). Nevertheless, a one-way flow of information could be enough to reach the visibility stage, as proposed by the Industry 4.0 maturity index (Dalenogare, Benitez, Ayala, & Frank, 2018; Frank, Dalenogare, & Ayala, 2019). This means that indicators from the production line can be combined with administrative information to support decision-making, e.g., to reorganize production orders based on the real efficiency of equipment, based on raw material purchasing schedule, or based on sales (Jaskó et al., 2020). In this kind of MES 4.0, besides the use of IoT systems for real-time data collection, other complementary technologies can be present, such as cloud systems for

integrating data across different processes and basic modules of big data analytics for data visibility.

The third configuration of MES 4.0 (Table 3) focuses on a complete vertical integration to leverage digital twins or autonomous processes (Cimino et al., 2019). The vertical integration concept of Industry 4.0 considers MES its core, connecting the manufacturing technology systems like sensors, machine-to-machine communication (M2M) protocols, supervisory control and data acquisition (SCADA) systems, and warehouse management systems (WMS), with corporate information systems like Material Requirement Planning (MRP), Enterprise Resource Planning (ERP) and Customer Relationship Management (CRM) systems. In this sense, MES 4.0 operates as the ‘brain’ connecting the factory’s operation with the other enterprise business processes (e.g., purchasing, human resources, product design, etc.) (Benitez et al., 2023). For such an intelligent vertical integration, information must have a two-way flow, providing operational real-time data to the corporate level as well as real-time resource planning for the operational level. This information two-way flow provides the condition for implementing digital twins and/or autonomous operations based on algorithms that adjust the smart manufacturing processes according to the defined variables (Mantravadi et al., 2023; Bueno et al., 2020). The IoT systems for real-time data collection will be integrated with cloud systems for sharing data across different processes. Also, big data analytics using AI for autonomous decision-making are frequently used, especially to set the production schedules autonomously, based on data analysis and simulations from the digital twins (Bueno et al., 2020).

The interviewees explained that several misunderstandings between buyers and providers happened due to the lack of understanding regarding customer needs or MES functionalities. Most customers did not understand the differences between MES 4.0 highlighted in Table 3. Therefore, they treated the technology as a single entity, leading to information asymmetry between actors in several cases. To address this issue, we selected three dyads, one for each configuration type identified, to analyze in-depth how information and knowledge exchange occurred between the actors and how it could be improved to avoid asymmetry and increase the chances of project success. The details of these cases will be presented in the following section.

4.2. Case studies for each MES configuration and knowledge-sharing dynamics

4.2.1. Dyad 1 – MES 4.0 for local visibility

DigitalizationCo, a startup specializing in IoT solutions, offers an MES system as one of its flagship products, providing essential features for real-time process visibility. AutomationCo, a multinational manufacturer of trailers, semi-trailers, and railcars, frequently invests in cutting-edge Industry 4.0 technologies and maintains a dedicated team of five professionals focused on integrating new technological solutions to enhance productivity.

In the preliminary stages of negotiation, AutomationCo approached DigitalizationCo with the intention of procuring the MES system to monitor critical equipment in real-time, encompassing functionalities such as data acquisition, quality management, process oversight, and performance analysis. Subsequently, DigitalizationCo proposed its pre-packaged MES kit, comprising IoT hardware for capturing equipment downtime and production volume, coupled with online software displaying key production management indicators like OEE, equipment downtime, and primary causes. The specifications of the MES kit were communicated to AutomationCo through a brochure available on the DigitalizationCo website. DigitalizationCo’s CEO recalled the interaction with AutomotiveCo personnel, stating, “*They requested additional indicators not included in the standard solution. I suggested starting with the fundamental indicators, and if needed, we could incorporate the additional ones later.*” Confident in their expertise, DigitalizationCo believed that the visibility sought by AutomationCo aligned with the initial stages of the Industry 4.0 maturity index (Frank et al., 2019), a requirement

consistent with those of other satisfied clients possessing similar characteristics to AutomationCo.

In the subsequent implementation phase, AutomationCo engaged DigitalizationCo for the comprehensive installation of the IoT kit for data collection. Initially, a proof-of-concept involved monitoring five pieces of equipment at the production site using the solution. When equipment downtime occurred, the system automatically logged the duration, prompting operators to input the reasons for the stoppage via a user-friendly human-machine interface (HMI). Similar procedures were followed for identifying and addressing quality issues. Leveraging this data, the software computed key performance indicators (KPIs) and presented them through online dashboards. AutomationCo utilized these dashboards to identify primary productivity challenges, investigate underlying causes, and spearhead continuous improvement initiatives. Following the successful proof-of-concept, the solution was rolled out to all critical equipment. The Digital Manufacturing Engineer expressed satisfaction with the solution, affirming, “*We now possess all necessary information through a solution that is remarkably easy to install and operate, all at a very competitive price point.*”

In this scenario, the buyer-provider configuration operated as a **black box**, wherein the technology solution was exclusively developed by the provider in a standardized manner, without the buyer's direct involvement (Le Dain & Merminod, 2014). To mitigate information imbalances through knowledge sharing, both parties needed to align expectations and comprehend the system requirements and the buyer's objectives. However, the MES remained unaltered, retaining its original features as provided by the vendor. This exemplifies a classic black box situation, where knowledge exchange primarily involves technical alignment from the buyer to the provider (Ayala et al., 2017).

In such instances, a ‘**knowledge transfer**’ dynamic suffices, particularly when the MES's objective is limited to achieving local visibility. As observed in this case, the complexity of this solution is minimal, and written documentation proves adequate since the buyer acquires the MES as a prepackaged solution with standard functionalities akin to those used by numerous companies. The primary challenge regarding information asymmetry lies in comprehending the system's functionalities and how they align with the local operational requirements. Given these requisites' narrow and specific nature, such as monitoring a production process, both parties must exchange technical details regarding the MES and the production process. This exchange typically relies on a shared lexicon of specifications concerning what the system offers and the production activity necessitates (Carlile, 2002). Hence, the focus lies on aligning technical understandings and requirements, facilitating effective communication and implementation. Based on these observations, we built the following proposition:

Proposition 1. *In instances where the MES configuration is designed primarily for local visibility objectives, a buyer-provider arrangement operating at the black box level, which entails knowledge transfer, could be adequate in mitigating information asymmetry.*

4.2.2. Dyad 2 – MES 4.0 one-way integration

SoftwareCo is a technology provider of software solutions developed for the furniture industry. It has consolidated solutions in the design, production, and management software market of several countries. The buyer, FurnitureCo, is a large company that manufactures highly customized furniture. This company stands out in the high-end furniture segment, serving customers in several countries. FurnitureCo constantly seeks to invest in technologies in its plant to increase production flexibility. In this scenario of high product customization and the consequent search for flexibility in the factory, FurnitureCo's Industrial Director explains that he seeks to implement Industry 4.0 to gain data visibility in real-time and with wireless communication. “I want real-time data visibility so I can know exactly what's happening in the factory at any given time”. Thus, a one-way vertical integration through MES implementation was considered to give the firm the aimed stage of Industry 4.0

maturity, i.e., factory visibility (Frank et al., 2019).

According to FurnitureCo's Industrial Engineer, the MES system was purchased from SoftwareCo because they were already a partner providing software for furniture design. FurnitureCo's established the main characteristics of the MES required for the factory and demanded it from SoftwareCo as provider. In this sense, the buyer-provider configuration employed in this case was a **white box** because the solution provided was completely developed according to the requisites established by the buyer (Le Dain & Merminod, 2014). This configuration was needed instead of a black box as in the previous case, considering that the integration aimed at the full factory with its characteristics and multiple processes that needed adaptation and customization from the system provided, resulting in a very specific solution. According to Ayala et al. (2017), this kind of configuration is usually associated with knowledge translation, meaning that the provider needs to translate the buyer's requisites into specific product deliveries. Different from the previous case in which standardized solutions are acquired that need an information alignment between the parties, this configuration demands a higher level of knowledge sharing, where interpretation and assimilation are required to translate such requisites into product characteristics (Carlile, 2002).

However, this particular project encountered several challenges stemming from the inadequate management of the knowledge translation process from the buyer to the provider. Upon analysis of the pre-contract phase, several issues contributing to information asymmetries were identified. FurnitureCo expressed interest in implementing an MES system, with SoftwareCo offering only a website link to an MES software tailored for furniture companies. SoftwareCo assured FurnitureCo that this MES solution would fulfill the requisite plant visibility needs, encompassing all requested functionalities such as complete machine integration, ERP integration, and additional modules for quality and maintenance processes. However, during negotiations, SoftwareCo provided only vague and superficial explanations of the MES functionalities. Influenced by the existing trust within the dyadic relationship and possessing limited knowledge of MES systems, FurnitureCo decided to proceed with the software purchase. In the post-contract phase, FurnitureCo demanded SoftwareCo to provide comprehensive plant visibility based on the vertical systems integration principle outlined by Industry 4.0. However, FurnitureCo's Head of Innovation expressed uncertainty regarding the software's scope, stating, “There's a widespread belief that the SoftwareCo system will resolve all plant issues with real-time data visibility, but nobody comprehends how it will happen.” SoftwareCo struggled to grasp FurnitureCo's needs and define how, or if, their software could meet those requirements. After months of misunderstandings between the parties, FurnitureCo engaged a management consultancy firm to precisely define and translate its needs and assess whether SoftwareCo's system could fulfill them. This consultancy delineated the architecture of the MES system, specifying the necessary functionalities, data sources, legacy system integration points, and operational procedures. Only upon finalizing the architecture did SoftwareCo realize that FurnitureCo's requirements exceeded the capabilities of its system. The SoftwareCo solution solely functioned as a production data viewer, lacking the expertise or infrastructure to directly extract data from production equipment. Ultimately, after a year of grappling with these challenges, FurnitureCo opted to terminate the contract and pursue a standardized MES solution with the requisite functionalities elsewhere.

Reflecting on the case study of the one-way configuration, it becomes evident that the MES was procured and supplied solely through a ‘**knowledge transfer**’ interaction between the involved parties. In this mode, both actors independently interpreted what an MES should entail, leading to numerous issues and an ultimately unsuccessful implementation. Given the complexity associated with the MES required for vertical integration within the Industry 4.0 framework, a more appropriate approach would have been ‘**knowledge translation**’, as emphasized by the system engineer from SoftwareCo: “*We only grasped*

what FurnitureCo truly needed after engaging in direct discussions facilitated by the consultancy company.” During this mediated meeting, MES functionalities were thoroughly deliberated, and the MES architecture was meticulously outlined. It was only through this collaborative process that both parties arrived at a shared understanding and common knowledge regarding the project requirements.

Based on this case, it becomes evident how crucial intermediaries are when knowledge translation is required. While the operation was configured to function in a white box model, it was necessary for someone to translate all the meanings and requirements from the buyer into clear specifications to be addressed by the provider. Intermediaries, such as the consultancy company in this instance, play a pivotal role in this process. They possess the ability to view problems from a broader perspective, unencumbered by routine issues, and are often better equipped to abstract knowledge and translate it into a different context (Frank and Ribeiro, 2014). Therefore, based on these insights, we propose the following:

Proposition 2. *When the objective of the MES configuration is one-way integration, a buyer-provider configuration operating at the white box level, which includes knowledge translation facilitated by intermediaries, can suffice to mitigate information asymmetry.*

4.2.3. Dyad 3 – MES two-ways integration

SystemCo, is a software company founded in 2005 that provides solutions for large-sized companies. According to SystemCo's Industrial Engineer, the company positioned itself early in the concept of Industry 4.0. “We started with Industry 4.0, big data, and other concepts before these terms even existed, and we've become well known today.” The buyer, AgriculturalCo, is a multinational manufacturer focused on developing, manufacturing, and distributing agricultural equipment to 140 countries and is headquartered in the US. Our case study at AgriculturalCo was carried out in a factory located in South America, which is advanced in the adoption of Industry 4.0 technologies. AgriculturalCo considered that implementing an MES for vertical integration of systems based on real-time data would be essential to continue with this journey towards the highest levels of Industry 4.0 maturity, usually associated with an autonomous factory (Frank et al., 2019).

AgriculturalCo directed its focus towards an MES tailored for two-way integration, as the company aimed to oversee production planning and control (PPC) through this system, aligning with the concept of smart PPC inherent in Industry 4.0 (Bueno et al., 2020). Smart PPC entails the ability to reorganize manufacturing resources and schedules based on the real capacity and condition of the manufacturing system (Bueno et al., 2020). This needs the integration of multiple informational layers and departments (production, planning, maintenance, quality, etc.), prompting the company to engage in close collaboration with providers to develop customized solutions that leverage knowledge from both sides. Consequently, this scenario characterizes a grey box configuration, recognizing the complexity inherent in both parties and striving to foster robust dialogue and cooperation to co-create the solution collaboratively (Le Dain and Merminod, 2014).

During the pre-contract phase, AgriculturalCo engaged in negotiations with SystemCo with the objective of procuring an MES to enhance production flow, detect faults, monitor Overall Equipment Effectiveness (OEE), and manage material movements and systems integration in alignment with the requirements for Smart PPC. The information provided by SystemCo regarding their MES solution resonated with the Industry 4.0 concepts of connecting the factory floor to the company's management systems, which aligned well with AgriculturalCo's expectations. Throughout the negotiation process, there was a concerted effort between the parties to establish a shared understanding of the MES architecture. SystemCo showcased its expertise in machine and system integration, communication development, and IoT-related technologies. Meanwhile, AgriculturalCo believed that SystemCo possessed the capabilities to fulfill its MES requirements through systems integration,

thereby enabling AgriculturalCo to embark on its Industry 4.0 journey.

In the post-contract phase, AgriculturalCo entrusted SystemCo with the development of the MES according to the defined functionalities required to meet the company's Industry 4.0 standards. Despite SystemCo's MES being touted as a competitive solution in the market with regular software updates geared towards Industry 4.0, AgriculturalCo found itself unable to fully utilize the system as promised during the pre-contract phase. As lamented by AgriculturalCo's IT manager during an interview, “It seems that SystemCo has an impressive solution, but we were only able to leverage about 10 % of its capabilities. We ended up using just a production display, which compares forecasted production with actual production and tracks losses over time.” According to AgriculturalCo's IT manager, one of the primary reasons for this failure was the system's lack of maturity and the inadequacy of the infrastructure to support its full functionality. This sentiment was echoed by the Business Analyst at SystemCo, who remarked, “We only grasped the complexity of the operation and the challenges posed by legacy systems after initiating the project [...] The full integration of systems is a very challenging task.”

After realizing that integrating all systems through the MES couldn't be simply delegated, both firms embarked on a collaborative effort to tackle the project. Teams comprising individuals from IT, OT, and process engineering at AgriculturalCo joined forces with SystemCo to develop the MES architecture collaboratively. However, beyond merely outlining the architecture, they recognized the need to create a roadmap outlining the necessary steps to achieve the desired full vertical integration essential for implementing the anticipated autonomous factory. These steps encompassed enhancing connectivity infrastructure to support all equipment, facilitating communication across various equipment protocols, and integrating all legacy systems used for data collection and analysis from equipment, among other critical aspects. As articulated by AgriculturalCo's IT Manager, “Clarity and progress only emerged once we began working together collaboratively.”

In this case, we observe that the buyer company initially engaged in a ‘knowledge translation’ dynamic, similar to Dyad 2. However, as the process unfolded, they had to shift their focus towards a ‘knowledge transformation’ dynamic. This transformation was necessitated by the realization that the MES software constituted just one component of the broader vertical integration journey. Given the complexity of the solution, neither the buyer nor the providers could independently develop the complete MES architecture. The knowledge transformation was evidenced by both sides needing to delve into the domain of the other. AgriculturalCo had to familiarize themselves with the capabilities of the MES, discerning what could feasibly be implemented to achieve smart PPC. This entailed reducing information asymmetries concerning MES specifications and the software team's skills in developing customized applications. Instead of merely purchasing the solution, AgriculturalCo fostered daily collaboration through round table discussions and cycles of field tests and adjustments in tandem with SystemCo. Conversely, SystemCo had to gain insights into the production process details and product characteristics of AgriculturalCo. They also had to adapt their MES architecture to align with AgriculturalCo's requirements, thereby narrowing the gap in understanding process specifications and even the company's culture to ensure the MES's effective integration into the workers' daily routines. This dynamic collaboration resulted in a prolonged yet fruitful process, culminating in profound and solid results for the Industry 4.0 journey. They attained one of the highest levels of digital maturity outlined in the Industry 4.0 roadmap, integrating real-time visibility with manufacturing activity planning and execution (Frank et al., 2019). Both parties acknowledged that this achievement was only possible due to their collaborative efforts, which facilitated the bridging of knowledge domains and a deeper comprehension of the production system's real problems and needs. Based on these observations, we propose the following proposition:

Proposition 3. *“When the MES configuration aims to achieve two-way integration, it is necessary to establish a buyer-provider configuration at the*

grey box level, which involves knowledge transformation, to prevent information asymmetry.”

4.3. Consolidation of the case study findings

Based on the insights collected from our multiple case study observations can be summarized. Firstly, it is evident that different buyer-provider relationships are necessary to develop MES configurations that align with buyers' needs within an Industry 4.0 context. Interestingly, as the complexity of the relationship grows between both parties, the size of the companies also increases. In Table 2 we provided the characteristics of the companies. When this is matched with the findings of the three dyads, summarized in this section in Fig. 2, it is possible to note that a small IoT company provided the solution for local visibility. Something similar happened when the implementation targeted a factory's visibility. However, a complete solution considering a smart PPC implemented through the MES 4.0 required high complexity and strong collaboration. In this case both companies of the dyad were large.

Second, regarding the collaboration, both parties must recognize that varying knowledge-sharing dynamics must be employed to mitigate information asymmetry. Fig. 2 summarizes these findings in a framework, presenting the increasing complexity of MES from a bottom-up perspective and the corresponding knowledge-sharing dynamics. Additionally, the increasing complexity of MES and knowledge sharing requires an evolution from a black box to a white box and, ultimately, to a grey box configuration. Our findings demonstrate a direct correlation between each buyer-provider configuration and the type of MES 4.0 demanded, as illustrated in Fig. 2. For instance, when the objective is as straightforward as monitoring local visibility, significant interventions in the MES may not be necessary. In such cases, a market-standardized solution can suffice, facilitating a black box mode of interaction with the technology provider. However, as complexity increases, the provision of the MES 4.0 solution needs to be more flexible and tailored to the specific requirements of the factory. This is particularly evident in the case of one-way integration MES types, where it becomes imperative to ensure that the system accommodates the diverse processes and activities that need to be integrated into a single cohesive system. Consequently, the MES solution is provided on-demand, aligning with the customer's specifications, thereby characterizing a white box configuration. Furthermore, the development of a two-way integration MES type demands the highest level of collaboration between the company and the provider. This collaboration is proposed in the form of a grey box collaboration, where both sides contribute their expertise and

knowledge domains to co-create the solution. Our research indicates that these levels of collaboration and types of solutions are intricately connected to the knowledge dynamics depicted in Fig. 2.

From the perspective of buyer-provider relationships, implementing an MES 4.0 presents significant challenges due to the complexity involved in integrating legacy systems (Calderón Godoy et al., 2018). As evidenced in the case studies, a mere knowledge transfer dynamic between the actors is insufficient. To mitigate information asymmetry effectively, both the buyer and the provider must engage in a 'knowledge translation' dynamic, utilizing the MES architecture as a boundary object (Carlile, 2002). On one side of this collaboration, the buyer must involve its IT team to present existing systems and hardware, while also engaging its manufacturing team to delineate the necessary modules of the MES required to achieve the desired visibility. On the other side, the provider must articulate the functionalities of its MES and any integration restrictions that may apply. As emphasized by Ayala et al. (2017b) and Le Dain & Merminod (2014), the complexity of this task cannot be understated and necessitates, at the very least, a white box buyer-provider configuration.

In this collaborative configuration, the buyer's clear understanding of its needs and the proposed solution's functionality is paramount before handing over the project to the provider for development. This ensures alignment and equips both parties with the necessary knowledge to navigate the complexities of implementing an MES 4.0 solution effectively. As observed in the case studies, achieving the highest maturity level of Industry 4.0 (Frank et al., 2019) demands a high level of complexity in MES configuration that can only be addressed through joint development between the buyer and the provider. This collaboration represents a grey box buyer-provider collaboration. During this collaboration, a 'knowledge transformation' occurs, as neither the buyer nor the provider possesses the complete knowledge of the final MES architecture. Instead, they work together to develop it based on the IT and OT demands. At this level, the complexity requires a tailor-made MES or extensive customization to integrate it with the buyer's existing software legacy and the various protocols of the equipment on the shop floor. This collaborative approach ensures that the MES solution meets the unique requirements of the buyer's operations while leveraging the expertise of both parties to achieve the desired outcome.

4.4. A decision-framework for MES 4.0 implementation

The observed dyads detailed in our findings have provided valuable insights into MES 4.0 adoption and the measures necessary to mitigate

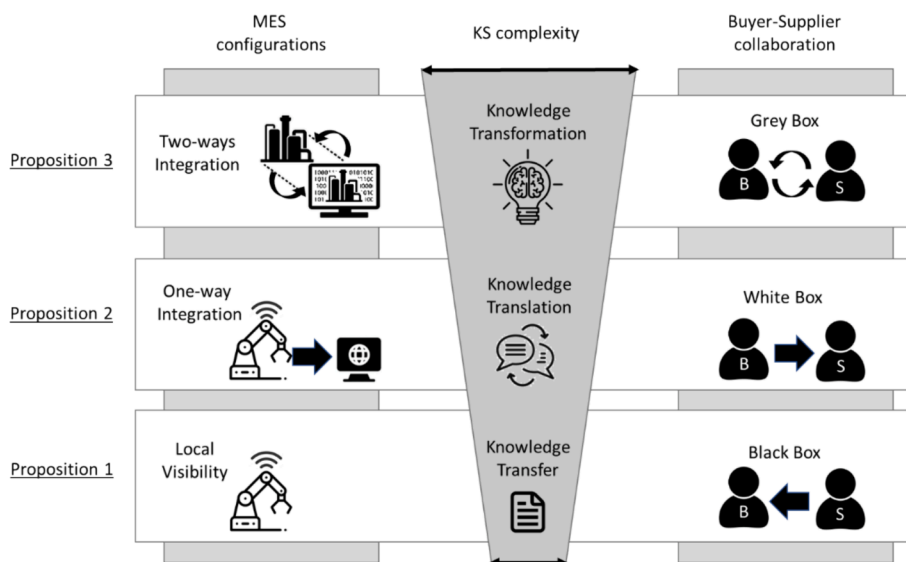


Fig. 2. Framework for MES functionalities, KS dynamics and Buyer-Provider collaboration.

information asymmetries between buyers and providers. We have translated this understanding into a solution to aid managers in the adoption decision process. We have devised a three-step decision framework for implementing MES 4.0, depicted in Fig. 3. This model can be examined in conjunction with a specific objective and the MES implementation for that purpose, thus aligning with the three steps. Alternatively, it can also be viewed from the perspective of MES 4.0 expansion, considering the vertical evolution in the model and expanding vertical integration through the MES's evolutionary process. We will discuss both approaches in the following sections.

Step 1 – Decision of the objectives for the MES 4.0

The first step involves defining the scope of what the company aims to achieve through the implementation of an MES in its Industry 4.0 journey. Our results indicate three possible options a) Monitoring Specific Production Indicators: This objective entails simpler solutions based on IoT monitoring systems. The focus is on supporting production line supervisors who need to visualize key indicators in their production line. b) Factory Visibility: This objective requires broader solutions to integrate different production processes performed in the factory into the same system. The goal is to understand the factory's capacity and performance as an integration of various production lines. c) Implementing a Smart Production Planning and Control: This objective goes beyond visualization and monitoring in real-time; it involves the execution of manufacturing activities through the MES solution. Here, the MES is more complex and functions as the real “brain” of the manufacturing process.

Step 2 – Decision of the type of MES 4.0 to be implemented

The second step involves the selection of MES 4.0 based on three types: local visibility, one-way integration, and two-way integration.

Fig. 3 illustrates how each of these MES types should be chosen according to the objective identified in Step 1. Additionally, it demonstrates that companies can progress in their adoption of MES based on the objectives of MES utilization. This progression is depicted as vertical integration steps in Step 2. For instance, a local visibility-oriented MES can evolve into a one-way integration MES, and eventually, into a two-way integration MES. However, if a company aims to progress in this direction by expanding its objectives, careful planning is necessary. Acquiring different solutions for each objective may prove to be too costly for the company. Therefore, decision-makers need to assess whether MES solutions available in their markets permit such evolution. In our case studies, we identified a two-way integration MES that allowed for starting with a basic IoT local solution and, through the addition of layers, achieving two-way integration. This evolution was made possible by expanding the MES functionalities during the evolutionary process. However, it is crucial that the MES 4.0 solution offers a modular structure, enabling the addition of new functionalities as objectives expand. This ensures adaptability and scalability in meeting the evolving needs of the company.

Step 3 – Managing knowledge sharing dynamics to reduce information asymmetries

Step 3 focuses on managing knowledge-sharing dynamics to mitigate information asymmetries between the buyer and the provider during the MES implementation. As depicted in Fig. 3, each type of MES 4.0 will necessitate distinct knowledge-sharing dynamics and forms of buyer-provider collaboration. In our findings, we outlined the requisites for these knowledge-sharing dynamics. Similarly, to the expansion of vertical integration through MES 4.0 outlined in Step 2, knowledge-sharing dynamics can evolve between the buyer and supplier—from knowledge

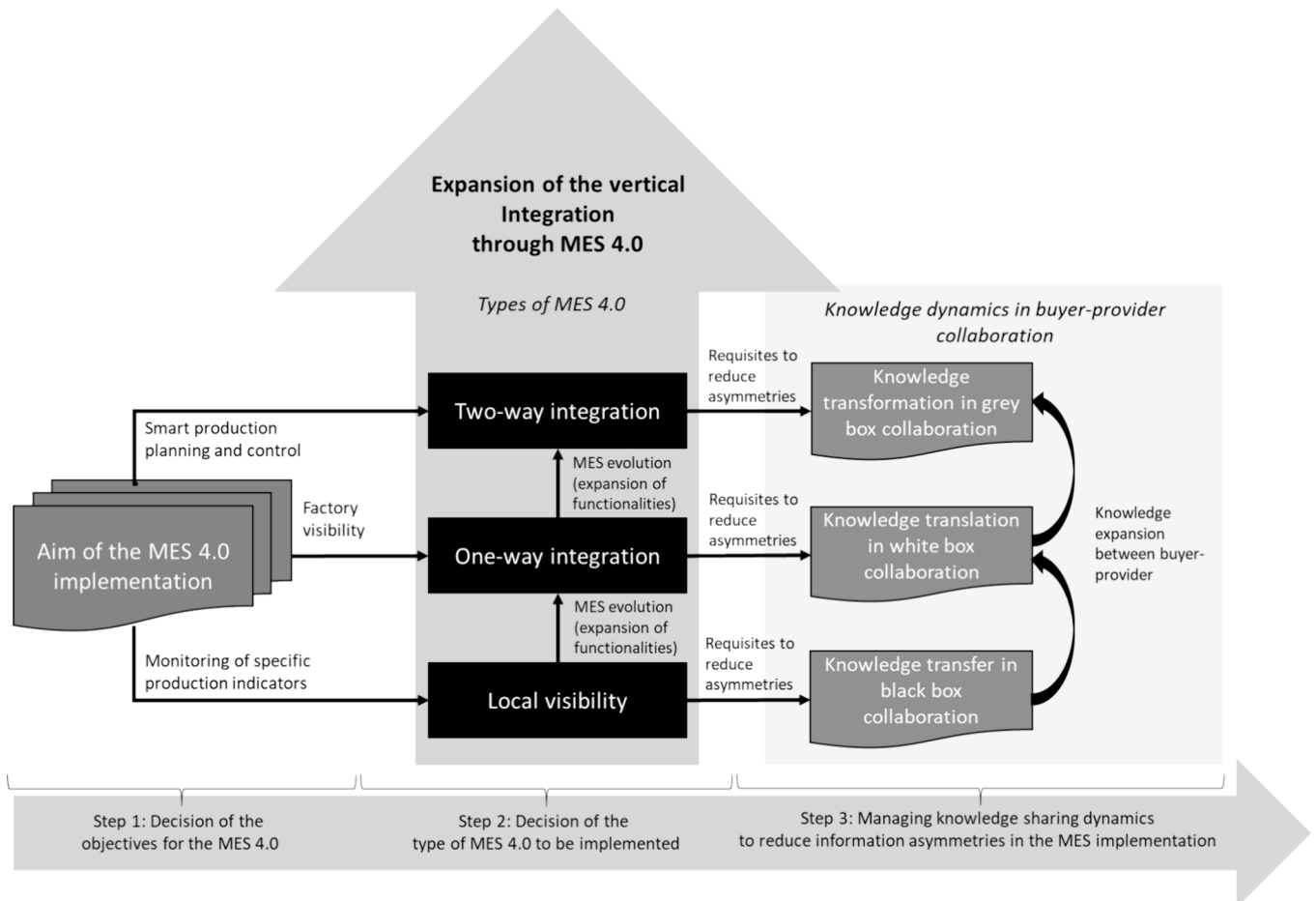


Fig. 3. Decision-framework for MES 4.0 implementation.

transfer to knowledge translation and ultimately to transformation. Meanwhile, collaboration between both sides becomes more transparent—progressing from a black box to a white box and eventually to a grey box. These dynamics will entail specific requisites to reduce information asymmetries, as discussed in the three dyads detailed in the results (Section 4.2).

5. Discussions and conclusions

5.1. Theoretical contributions

Our study contributes significantly to the existing body of theory in several ways. Firstly, while the information systems literature has extensively examined technical and organizational factors influencing MES adoption, our study represents the first comprehensive qualitative exploration of MES implementation within the context of Industry 4.0. This involves an MES that integrates various technologies such as automation technologies, sensors, SCADA, and ERP to facilitate vertical integration and real-time data streaming based on IoT. The novelty of this perspective lies in considering an interrelated and complex system of different information technologies, which necessitates a broader set of requirements than studying the adoption of a single information system in isolation. In this sense, one important contribution is that we provide three types of MES 4.0, related to different levels of technological complexity. We provide a future perspective on the role of MES 4.0 when we introduce the two-way integration of MES 4.0. New technologies are included in this type of system, especially digital twins. Prior research has pointed out opportunities for the integration between MES and digital twins (Nakagawa et al., 2021; Negri et al., 2020), and we expand this concept by showing that the most advanced level of MES 4.0 should be built around digital twins using vertical integration as a background. Benitez et al. (2023) considered both MES and digital twins as different Industry 4.0 platforms that can help interconnect different technologies to create a smart operation. However, based on our findings, they should be seen as a unique integrated platform that operates based on a digital model of the factory with real-time data that allows organizing the manufacturing execution. The future evolution of MES will demand architectures for better integration with digital twins (Leng et al., 2023).

Moreover, our study highlights how technological complexity can lead MES buyers and sellers to interpret system characteristics differently. By employing the information asymmetry lens (Akerlof, 1978), we illustrate that successful MES purchasing and implementation for Industry 4.0 begins from the pre-contracting stage—a stage often underestimated or overlooked by both academics and practitioners. Furthermore, while prior studies have examined MES implementation challenges in Industry 4.0 from a static transactional perspective, our study delves into the dynamic aspects of implementation that information systems managers need to consider. This shift in focus offers valuable insights into the evolving nature of MES implementation and presents opportunities for future research to explore the socio-technical aspects of adopting Industry 4.0 technologies in greater depth. Lastly, by integrating perspectives from knowledge sharing (Carlile, 2004) and buyer-provider configuration (Petersen et al., 2003), we demonstrate that different levels of MES complexity require varying types of knowledge-sharing dynamics to bridge the gap between actors, as well as different types of buyer-provider collaboration. This synthesis of theories has culminated in a conceptual framework that scholars can adapt to analyze the adoption of complex information systems across various contexts.

5.2. Managerial contributions

In previous research, Tabim et al. (2021) observed that the same provider of MES was found to be suitable for one buyer while being inadequate for another. This discrepancy arose because the buyers' needs differed, yet the provider offered the same software configuration

and service. Thus, although MES are marketed as a singular product, they actually come with various configurations of functionalities. Consequently, our study can serve as a guiding resource for buyers and providers when deciding on MES implementation as a gateway to Industry 4.0.

Firstly, small firms with limited human and financial resources for investing in Industry 4.0 technologies can find off-the-shelf solutions in the market. These solutions typically offer only basic MES functionalities and plug-and-play hardware, providing equipment visibility sufficient for productivity improvement at affordable prices (Mittal et al., 2018).

Secondly, more mature firms aiming for vertical integration need to internally develop knowledge on how the MES functions and how its functionalities and systems will integrate. This requires a specialized internal IT team to oversee the MES development outsourced to the provider. However, this may pose a significant barrier for manufacturing firms in achieving an advanced level of Industry 4.0, particularly if they lack a well-developed IT team, which is often the case (Marcucci et al., 2022).

Thirdly, firms already advanced in their Industry 4.0 journey and aiming for the advanced level with 'live' digital twins or autonomous processes need to delve deeper into their MES and vertical integration implementation. However, due to the complexity and innovative nature of this MES for Industry 4.0, off-the-shelf purchases or internal development of its architecture are not viable options. Success in this scenario hinges on finding a provider willing to develop a customized MES solution tailored to integrate the buyer's systems completely. Additionally, the buyer must have well-prepared IT and OT teams familiar with the firm's current systems and the requirements for achieving desired digital twins and autonomous processes.

In conclusion, our study collaborates with practitioners to ensure success on their Industry 4.0 journey. Many manufacturing firms face challenges in their MES implementation projects due to inadequate internal capabilities or unclear expectations from the implementation. Likewise, many MES providers struggle in their relationships with buyers, failing to meet expectations not due to a lack of capacity but because of information asymmetry. Thus, our study provides support and insight into the different types of MES implementations and the knowledge-sharing dynamics necessary for a successful buyer-provider relationship.

5.3. Limitations and future research

As is common in academic research, while rigorous methodological procedures were adhered to, this study has limitations. Firstly, the study only examined three case studies, which provided sufficient insight into different levels of MES complexity and knowledge-sharing dynamics. However, future research could enhance the study's breadth by analyzing additional case studies and exploring technologies beyond MES. Secondly, researchers can apply the framework proposed in Fig. 2 and the resulting guideline presented in Fig. 3, which encompasses MES functionalities, knowledge-sharing dynamics, and buyer-provider collaboration, to other technologies. If the framework's generalizability is established, it could serve as a valuable guide for IT and OT managers during the procurement and implementation processes of various Industry 4.0 technologies. Moreover, due to the integration of systems demanded by Industry 4.0, the functionalities of MES increasingly overlap with those of other systems, such as ERP or PLM. This convergence may lead to misunderstandings between buyers and providers. Hence, future research could delve deeper into the distinct roles of each system in this new Industry 4.0 era. In addition to the aforementioned limitations and areas for future research, it is worth considering the importance of top management support and resistance to change in system integration, like usually happens for ERP (Gunjal, 2019). Exploring information asymmetry and top management support in knowledge sharing could yield valuable insights into addressing these challenges effectively.

CRediT authorship contribution statement

Verônica M. Tabim: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft. **Néstor F. Ayala:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft. **Giuliano A. Marodin:** Validation, Writing – review & editing. **Guilherme B. Benitez:** Formal analysis, Investigation, Methodology. **Alejandro G. Frank:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgements

The authors thank the Brazilian National Council for Scientific and Technological Development (CNPq – Conselho Nacional de Desenvolvimento Científico e Tecnológico) (CNPq PQ1D, PQ2, and CNPq DAI), and, the Research Council of the State of Rio Grande do Sul (FAPERGS, Fundação de Amparo à Pesquisa do Estado do Rio Grande do Sul) (Process no. 21/2551-0002161-6) for the financial support received to conduct this research.

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